Chapter One

Rational and Irrational Numbers

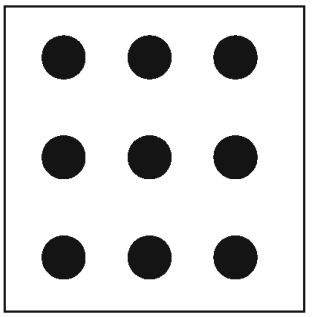
We quantify the objects around us by numbers. In our previous classes we have concepts of natural numbers, integer and fractional numbers. These numbers are known as rational numbers. These numbers can be expressed as the ratio of two integers. Apart from these, there are numbers which can not be expressed as the ratio of two integers and these numbers are known as irrational numbers. In this chapter we shall be acquainted with irrational numbers and their applications.

At the end of this chapter, the students will be able to–

- Explain square and square roots of a number.
- Find square roots by the methods of factorization and division.
- Solve real life problems by applying the methods of ditermination of square roots.
- Identify rational and irrational numbers.
- Locate the rational and irrational numbers in the number line.

1.1 Squares and square roots

Geometrically a square is a rectangle with equal sides. The square with a side of x units has an area x^2 square units. Conversely, if the area of a square is x^2 square units, then the length of each side is x units.



In the figure, 9 marbles are arranged in a square array. The marbles are placed at equal distances with 3 marbles in each of 3 rows. So the total number of marbles is $3 \times 3 = 3^2 = 9$. Here, the number of marbles in a row and the number of rows are equal. Hence, the figure is a square. We say that the square of 3 is 9 and the square root of 9 is 3.

∴ **A square is the product of multiplication of a number by itself and the number is the square root of the product.**

$2 \times 2 = 2^2 = 4$
(The square 2 is 4)
The square root of 4 is 2

1.2 Integer

Observe the following table :

Length of a side of the square (m)	Area of the square (m^2)
1	$1\times1=1=1^2$
2	$2\times2=4=2^2$
3	$3\times3=9=3^2$
5	$5\times5=25=5^2$
7	$7\times7=49=7^2$
a	$a\times a=a^2$

The characteristic of the numbers 1, 4, 9, 25, 49 is that they can be expressed as square of any other integer. Such numbers are square numbers.

The square root of a perfect square number is a natural number. For example, the square of 21 is 21^2 or 441 which is a perfect square and square root of 441 is 21 which is a natural number.

Generally, if a natural number m can be expressed as square (n^2) of another natural number n, then m is square number. Here the number m is known as a perfect square number.

Properties of Square Numbers

The square of numbers from 1 to 20 have been given in the following rows, Fill up the vacant boxes :

Number	Sq. number	Number	Sq. number	Number	Sq. number	Number	Sq. number
1	1	6	36	11	121	16	256
2	4	7	☐	12	☐	17	289
3	9	8	64	13	169	18	324
4	☐	9	81	14	196	19	361
5	25	10	☐	15	☐	20	☐

Let us observe the digits in the units place of the square numbers included in the table. It is to be noted that the digits in the units place of these number are 0, 1, 4, 5,6 or 9 only. The digits 2, 3, 7 or 8 are not in the units place of any square numbers.

Activity :

1. Will it be a square number if any number has any of the digits 0,1,4,5,6,9 in its unit place?
2. Which of the followings numbers are perfect square?
   2062, 1057, 23453, 33333, 1068.
3. Write five numbers whose digits in the unit place help to draw conclusion that they are not square numbers.

Let us consider square numbers having 1 in the unit place from the above table :

Square number	Number
1	1
81	9
121	11
361	19

If the digit in unit place is either 1 or 9, the digit in unit place of the square number will be ?

Similarly

Square number	Number
9	3
49	7
169	13

If the digit in unit place is either 3 or 7, the digit in unit place of the square number will be ?

and

Square number	Number
16	4
36	6
196	14
256	16

If the digit in unit place is 4 or 6, the digit in unit place of the square number will be ?

- The number consisting of digit 2 or 3 or 7 or 8 at the extreme right, that is, in the unit place can never be a perfect square.
- If odd number of zeros are in the right of a number, it will not be a perfect square.
- A number may be a perfect square if the digit at its unit place is 1 or 4 or 5 or 6 or 9. For example, 81, 64, 25, 36, 49 etc. are perfect square.

- Again, if even number of zeros are at the right of a number, the number may be a perfect square. For example, 100, 4900 etc. are perfect squares.

Activity :
1. From the table construct a rule for square numbers whose digit in the unit place is 4.
2. What would be the digit in the unit place of the following numbers : 1273, 1426, 13645, 9876474, 99580.

A table of a few perfect square along with their square roots are given below :

Sq. number	Sq. root	Sq. number	Sq. root	Sq. number	Sq. root
1	1	64	8	225	15
4	2	81	9	256	16
9	3	100	10	289	17
16	4	121	11	324	18
25	5	144	12	361	19
36	6	169	13	400	20
49	7	196	14	441	21

Symbol of Square Root

To express a square root, $\sqrt{\ }$ symbol are used. The square root of 25 is written as $\sqrt{25}$.

We know, $5 \times 5 = 25$, so the square root of 25 is 5.

Activity : Make a list of perfect squares from a few natural numbers.

Finding square root through prime factorization

Resolving 16 into prime factors we get

$16 = 2 \times 2 \times 2 \times 2 = (2 \times 2) \times (2 \times 2)$

Taking one factor out of every pair of factors, we get $2 \times 2 = 4$

$\therefore$ Square root of 16 = $\sqrt{16} = 4$

Again, resolving 36 into prime factors, we get

$$\begin{array}{r|l} 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline & 2 \end{array}$$

$36 = 2 \times 2 \times 3 \times 3 = (2 \times 2) \times (3 \times 3)$

Taking one factor but of each pair of factors,

we get $2 \times 3 = 6$

$\therefore$ Square root of 36 = $\sqrt{36} = 6$.

2	36
2	18
3	9
	3

Observe : To determine the square root of a perfect square with the help of prime factors–

- At first, the given number is to be resolved into its prime factors.
- Each pair of same factors has to be written together, side by side.
- One factor is to be written from each pair of factors of same type.
- The successive multiplication of the written factors will be the required square root.

Example 1. Find the square root of 3136.

Solution :

2	3136
2	1568
2	784
2	392
2	196
2	98
7	49
	7

Here, 3136 $= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 7 \times 7$

$= (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (7 \times 7)$

$\therefore$ Square root of 3136 = $\sqrt{3136} = 2 \times 2 \times 2 \times 7 = 56$

Activity : Determine the square root of 1024 & 1849 with the help of prime factors.

1.3 Finding the square root by division method

An example illustrates the method for finding the square root of a number :

Example 2. Find the square root of 2304 by division method.

Solution :

(1) Write down the number 2304: 23 04

(2) From right side, take two digits at a time and form pair. Place a horizontal line over every pair.

$\overline{23}\ \overline{04}$

(3) Draw a vertical line to the right of the number as in division.

$\overline{23}\ \overline{04}\ |$

(4) The first pair is 23. Its previous square number is 16, whose square root is $\sqrt{16}$ or 4. Write 4 at the right side of vertical line. Now write 16 just below 23.

$\overline{23}\ \overline{04}\ |4$
16

(5) Now subtract 16 from 23.

$\overline{23}\ \overline{04}\ |4$
16
7

(6) To the right of the result 7, put the next pair 04. Draw a vertical line (sign of division) to the left of 704.

$\overline{23}\ \overline{04}\ |4$
16
$|\ 7\ 04$

(7) Put twice of the quotient 4 i.e. (4×2) or 8 to the left side of vertical line. Keep a space for inserting a digit in between 8 and the vertical line.

$\overline{23}\ \overline{04}\ |4$
16
$8\ |\ 7\ 04$
$|$

(8) Now, find an one digit number which is to be placed at the right of 8 so that the number so formed when multiplied by that digit equals to 704 or less than704. In this case it is 8. Put this 8 to the right of 4 in the quotient.

$\overline{23}\ \overline{04}\ |48$
16
$88\ |7\ 04$
$7\ 04$
0

(9) 48 is obtained in the quotient. This is the required square root.

$\therefore\ \sqrt{2304} = 48$

N.B. : To find the square root by division method, if the last digit can not be in pair of forming pairs from right side, it has to be considered without pair.

Example 3. Find the square root of 31684 by division method.

Solution :

$$\begin{array}{r|l} 3\ \overline{16}\ \overline{84} & 178 \\ 1 & \\ \hline 27 \mid 216 & \\ 189 & \\ \hline 348 \mid 2784 & \\ 2784 & \\ \hline 0 & \end{array}$$

$\therefore$ Square root of 31684 = $\sqrt{31684}$ = 178

The required square root is 178.

Activity :

1. Find the square root of 1444 and 10404 by of division method.
2. Find the digits at units place of the square root of the numbers 529, 3925, 5041 and 4489.

Some important points on square and square root

- If a point is put on after every alternate digits of a number starting from right to the left, the number of points will be same as the number of digits in the square root.

For example,

$\sqrt{81}$ = 9 (consists of one digit, here number of dot over the number is one since $81 = \dot{8}\dot{1}$)

$\sqrt{100}$ = 10 (consists of two digits, here number of dots over the number are two since $100 = \dot{1}0\dot{0}$)

$\sqrt{47089}$ = 217 (consists of three digits, here number of dots over the number are three since $47089 = \dot{4}7\dot{0}8\dot{9}$)

Activity :

2. Find the number of digits of square roots of the numbers 3136, 1234321 and 52900.

Problems of square and its related

Example 4. What is the least number which is to be subtracted from 8655 to get a perfect square number ?

Solution :

$$\begin{array}{r|l} \overline{86}\,\overline{55} & 93 \\ 81 & \\ \hline 183 \quad 5\,55 & \\ \quad\;\; 5\,49 & \\ \hline \quad\quad 6 & \end{array}$$

Here, 6 is the remainder in finding the square root of 8655 by division.

Therefore, if 6 is subtracted from the given number, then the number will be a perfect square.

The required least number is 6.

Example 5. What is the least number which is to be added to 651201 to get a perfect square number ?

Solution :

$$\begin{array}{r|l} \overline{65}\,\overline{12}\,\overline{01} & 806 \\ 64 & \\ \hline 1606 \quad 1\,12\,01 & \\ \quad\;\; 96\,36 & \\ \hline \quad\quad 15\,65 & \end{array}$$

Hence, the remainder is 1565 in finding the square root. So the given number is not perfect square. The least number when added to 651201 will make the total sum a perfect square and then its square root will be 806 + 1 = 807

Square of 807 is $807 \times 807 = 651249$.

The required least number is 651249 – 651201 = 48.

Exercise 1.1

1. Find the square root of each by prime factorization:

   (a) 169 (b) 529 (c) 1521 (d) 11025

2. Find the square root of each by division method:

   (a) 225 (b) 961 (c) 3969 (d) 10404

3. What is the least number which is to be multiplied with the following numbers to get a perfect square number ?

   (a) 147 (b) 384 (c) 1470 (d) 23805

4. What is the least number which is to be divided by the following number so that the quotient would be a perfect square ?

   (a) 972 (b) 4056 (c) 21952

5. What is the least number which is to be subtracted from 4639 so that the difference is a perfect square ?

6. What is the least number to be added to 5605 so that the total sum is a perfect square ?

1.4 Finding square root of decimal fraction

The way in which the square root of the perfect square number or whole number is determined by long division, the square root of the decimal fraction is also determined in the same way. There are two parts of a decimal fraction. The part on the left side of decimal point is the whole or integral part and the part on the right side of decimal point is called decimal part.

Steps for finding a square root

(1) In the whole part, horizontal bar is to be drawn on two digits each from the unit place gradually to the left.

(2) In the decimal part, horizontal line is to be drawn over the digits in pairs from the right side of decimal point. If a digit is left alone in this way, then a zero is put beside the digit and the bar is put on two digits.

(3) In the usual way of determining square root, the activity over the integer part is carried out and a decimal point should be put in the square root before considering the first two digits after decimal point.

(4) For each pair of zeros in the decimal of a number, one zero is to be put after decimal point in the square root.

Example 1. Find the square root of 26·5225.

Solution :

$$\begin{array}{r|l} \overline{26}\cdot\overline{52}\overline{25} & 5\cdot15 \\ 25 & \\ \hline 101 \quad 1\,52 & \\ 1\,01 & \\ \hline 1025 \quad 51\,25 & \\ 51\,25 & \\ \hline 0 & \end{array}$$

The required square root = 5·15

Example 2. Find the square root of 0·002916

Solution :

$$\begin{array}{r|l} \overline{0}\cdot\overline{002916} & 0\cdot054 \\ 25 & \\ \hline 104 \quad 416 & \\ 416 & \\ \hline 0 & \end{array}$$

The required square root = 0·054

Determination of square root in approximate value

To find the square root correct upto three decimal places, at least 6 digits after the decimal are to be taken. If needed, after the last digit, zero is to be added to the right as required. It does not change the value of the number.

Example 3. Find the square root of 9·253 upto three decimal places. (approximate)

Solution :

$$\begin{array}{r|l} \overline{9}.\overline{25}\ \overline{30}\ \overline{00}\,\overline{00} & 3\cdot0418 \\ 9 & \\ \hline 604 \quad 25\,30 & \\ 24\,16 & \\ \hline 6081 \quad 1\,14\,00 & \\ 60\,81 & \\ \hline 60828 \quad 53\,19\,00 & \\ 48\,66\,24 & \\ \hline 4\,52\,76 & \end{array}$$

The required square root = 3·042 (approx.)

Example 4. Find the square root of 123 upto three decimal places. (approximate)

Solution :

$$\begin{array}{r|l} \overline{123}.\overline{00}\ \overline{00}\ \overline{00} & 11.090 \\ 1 & \\ \hline 21 \quad 23 & \\ 21 & \\ \hline 22\,09 \quad 2\,00\,00 & \\ 1\,98\,81 & \\ \hline 1\,19\,00 & \end{array}$$

The required square root = 11·090 (approx.)

N.B. : In the above square root, the fourth digit after decimal is being 8, 1 is added with third digit and the required value of square root (upto 3 decimal places) becomes 3·042.

Rules for finding approximate value of square root

- To find the square root correct upto two decimal places, the square root upto three decimal point is to be determined.
- If the next digit after decimal place upto which square root is to be determ ne is 0, 1, 2, 3 or 4, 1 should not be added with the previous digit.
- If the next digit after decimal place upto which square root is to be determined is 5, 6, 7, 8 or 9, 1 is to be added to the previous digit.

Activity:
1. Find the square root of 50·6944.
2. Find the square root of 7·12 upto two decimal places.

1·5 Perfect square fraction

If $\frac{50}{32}$ is reduced to least form, we get $\frac{25}{16}$

Here, the numerator of the fraction $\frac{25}{16}$ is 25, which is a perfect square number and denominator 16 is also a perfect square number.

So, $\frac{25}{16}$ is a perfect square fraction. ∴ When the numerator and denominator of a fraction are perfect square or numerator and denominator of a reduced fraction are perfect square, the fraction is said to be a perfect square fraction.

1·6 Square root of a fraction

The square root of a fraction is determined by dividing the square root of numerator by the square root of denominator of the fraction.

Example 5. Find the square root of $\frac{64}{81}$.

Solution : Square root of the numerator 64 of the fraction $= \sqrt{64} = 8$

and square root of denominator 81 $= \sqrt{81} = 9$

∴ Square root of $\frac{64}{81} = \sqrt{\frac{64}{81}} = \frac{8}{9}$

The required square root $= \frac{8}{9}$

Example 6. Find the square root of $52\frac{9}{16}$.

Solution : Square root of $52\frac{9}{16} = \sqrt{52\frac{9}{16}} = \sqrt{\frac{841}{16}} = \frac{29}{4} = 7\frac{1}{4}$

$\therefore$ Square root of $52\frac{9}{16} = 7\frac{1}{4}$

If the denominator of an fraction is not a perfect square number then it is to be transformed into perfect square by multiplication.

Example 7. Find the square root of $2\frac{8}{15}$ upto three decimal places.

Solution : The square root of $2\frac{8}{15}$

$$= \sqrt{2\frac{8}{15}} = \sqrt{\frac{38}{15}} = \sqrt{\frac{38\times15}{15\times15}}$$

$$= \sqrt{\frac{570}{225}} = \frac{23\cdot8747}{15} = 1\cdot5916 \text{ (app.)}$$

$\therefore$ The square root upto three decimal places = 1·592 (approx.)

Activity :

1. Find the square root of $27\frac{46}{49}$.
2. Find the square root of $1\frac{8}{5}$ upto two decimal places.

1·7 Rational and irrational numbers

1,2,3,4, ... etc. are natural numbers. These numbers can be expressed in the form of a fraction of two natural numbers.

$1 = \frac{1}{1}, 2 = \frac{2}{1}, 3 = \frac{3\times2}{2} = \frac{6}{2}$, ... etc.

Again, 0.1, 1·5, 2·03, ... etc. are decimal numbers.

Here, $0\cdot1 = \frac{1}{10}$, $1\cdot5 = \frac{15}{10}$, $2\cdot03 = \frac{203}{100}$ which are fractional forms of the numbers.

Again, $0 = \frac{0}{1}$, is a fractional number.

The numbers discussed above are rational numbers.

So, zero, all natural numbers and fractions are rational number.

Irrational Numbers : $\sqrt{2} = 1 \cdot 4142135 \ldots\ldots$. the numbers of digits after decimal are not fixed. So, it cannot be expressed in a fractional form of two natural numbers. Similarly, the number $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$ .... etc. can not be expressed in a fractional form of two natural numbers. These are irrational numbers.

Observe : $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$ ..... etc. are irrational numbers and the numbers 2,3,5,6, ... etc. are not perfect squares. So, the square root of the numbers which is not perfect square is an irrational number.

Example 8. Choose the irrational number from the following numbers

$0 \cdot 12, \sqrt{25}, \sqrt{72}, \frac{\sqrt{49}}{7}$.

Solution : Here, $0 \cdot 12 = \frac{12}{100} = \frac{3}{25}$, which is a fractional number.

$\sqrt{25}, = \sqrt{5^2} = 5$, which is natural number

$\sqrt{72} = \sqrt{2 \times 36} = \sqrt{2 \times 6^2} = 6\sqrt{2}$, which can not be written as fraction

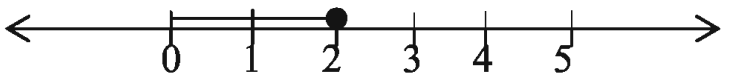
and $\frac{\sqrt{49}}{7} = \frac{\sqrt{7^2}}{7} = \frac{7}{7} = 1$, which is a natural number

$\therefore$ $0 \cdot 12, \sqrt{25}, \frac{\sqrt{49}}{7}$ are rational numbers and $\sqrt{72}$ is an irrational number.

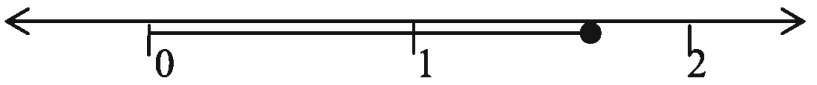
Activity :
1. Separate the rational and irrational number from $1\frac{1}{2}, \sqrt{\frac{4}{25}}, \sqrt{\frac{27}{16}}, 1 \cdot 0563, \sqrt{32}, \sqrt{121}$.

1·8 Expression of rational and irrational numbers in a number line Rational numbers on the number line

Observe the number line below :



The dark point on the above number line denotes the position of 2.



Again, in the above number line, the position of the dark point lies between 1 and 2. The denoted dark point lies at 3 out of 4 parts. Hence, the dark point denotes $1 + \frac{3}{4}$ or $1\frac{3}{4}$.

Irrational numbers on the number line

$\sqrt{3}$ is an irrational number where $\sqrt{3} = 1.732 \ldots\ldots = 1.7$ (approx.)

Now, dividing the segment in between 1 and 2 into 10 equal parts, mark the 7th part with a dark point which denotes 1·7. i.e. it denotes $\sqrt{3}$ approximately.



So, the darkened point is the location of of $\sqrt{3}$ on the number line.

Activity : 1. Locate the numbers 3, $\frac{3}{2}$, 1.455 and $\sqrt{5}$ on the number line.

Example 9. In a garden, there are 1296 mango trees. Along the length and breadth of the garden there are equal number of mango trees. Find the number of trees in each row of the garden.

Solution : There are equal number of mango trees in each row along both length and breadth of the garden.

$\therefore$ The number of trees in each row will be the square root of 1296.

Here,

$$\begin{array}{r|l} \overline{12}\ \overline{96} & 36 \\ 9 & \\ \hline 66 \quad 3\,96 & \\ 3\,96 & \\ \hline 0 & \end{array}$$

The required number of trees is 36.

Example 10. A scout team can be arranged in 9, 10 and 12 rows. Again, they can be arranged in a square form. Find the minimum number of scouts in that scout team.

Solution : The scout team can be arranged in 9, 10 and 12 rows. Therefore, the number of scouts is divisible by 9, 10 and 12. This least number will be L.C.M. of 9, 10 and 12.

Here,

2	9, 10, 12
3	9, 5, 6
	3, 5, 2

$\therefore$ L.C.M. of 9, 10 and 12 $= 2 \times 2 \times 3 \times 3 \times 5 = (2 \times 2) \times (3 \times 3) \times 5$

But obtained L.C.M. $(2 \times 2) \times (3 \times 3) \times 5$ can not be arranged in square form.

To make a perfect square $(2 \times 2) \times (3 \times 3) \times 5$ is to be multiplied at least by 5.

$\therefore$ The number required to arrange in 9, 10 and 12 rows and also in square form is $(2 \times 2) \times (3 \times 3) \times (5 \times 5) = 900$

The required number of scouts is 900.

Example 11. 21952 and 5605 are two numbers.

(A) Give reason whether the first number be perfect square number.

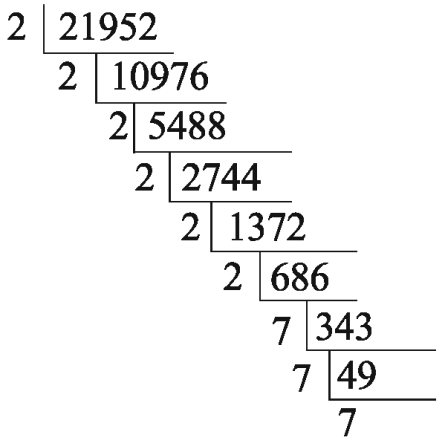
(B) If the first number is not perfect square number, what is the least number by which it is divided to get a perfect square number?

(C) What is the least number to be added to the second number so that total sum is a perfect square number?

Solution-1: (A) The number consisting of digit 2 or 3 or 7 or 8 at the extreme right that is, in the unit place can never be a perfect square. As the number 21952 has digit 2 in its unit place, the number is not perfect quare.

(B)

Here,

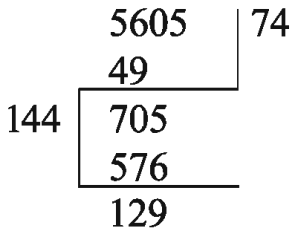


So, 21952= 2×2 ×2×2×2×2×7×7×7

The number 21952 is not perfect square. If we devide the number by 7, the gained number will be perfect square

Ans : 7

(C) Here,



As the remainder is 129 in finding the square root, the given number is not perfect square. The last number where added to 5605 will make the total sum a perfect square and then its square roof will be 74+1=75

Square of 75 = 75 × 75=5625

So, the required least number =5625−5605=20

Ans : 20

Exercise 1·2

1. Which one of the following is the square root of $\frac{289}{361}$?

   (a) $\frac{13}{19}$ (b) $\frac{17}{19}$ (c) $\frac{19}{13}$ (d) $\frac{19}{17}$

2. Which one of the following is the square root of 1.1025 ?

   (a) 1.5 (b) 1.005 (c) 1.05 (d) 0.05

3. A rational number is-

   (i) 0

   (ii) 5

   (iii) $\frac{5}{2}$

   Which of the following is correct?

   (a) i & ii (b) i & iii (c) ii & iii (d) i, ii & iii

   The difference of squares of two consequtive numbers is 19.
   Answer to question no. 4 and 5 following the information.

4. If one number is 10, what is the other number?

   (a) 12 (b) 11 (c) 9 (d) 8

5. what is the sum total of squares of the two numbers?

   (a) 281 (b) 221 (c) 181 (d) 164

6. which of the following is the square root of 0.01?

   (a) 0.01 (b) 0.1 (c) 0.2 (d) 1

7. If the digit in unit place of a number is either 2 or 8, the digit in unit place of its square will be –

   (a) 2 (b) 4 (c) 6 (d) 8

8. By which number the multiplication or division of $3\times7\times5\times7\times3$ will be perfect square number?

(a) 3 (b) 5 (c) 7 (d) 11

9. Which one of the irrational number ?

(a) $\sqrt{2}$ (b) $\sqrt{9}$ (c) $\sqrt{16}$ (d) $\sqrt{25}$

10. A farmer buys 595 plants for making a garden. The price of each plant is Tk. 12

(a) How much money did he spend to buy the plants?

(b) How many of the plants will be left if number of plants in each row of the garden is equal to number of rows?

(c) What is the least number which is to be added to the difference of the number of spending of total taka and the number of plants so that the sum will be a perfect square number?

11. Find the square root:

(a) 0·36 (b) 2·25 (c) 0·0049 (d) 641·1024
(e) 0·000576 (f) 144·841225

12. Find the square root upto two decimal places:

(a) 7 (b) 23·24 (c) 0·036

13. Find the square root of the following fractions:

(a) $\frac{1}{64}$ (b) $\frac{49}{121}$ (c) $11\frac{97}{144}$ (d) $32\frac{241}{324}$

14. Find the square root upto three decimal places:

(a) $\frac{6}{7}$ (b) $2\frac{5}{6}$ (c) $7\frac{9}{13}$

15. At least how many soldiers is to be removed or is to be added with 56728 soldiers so that the soldiers can be arranged in form of a square?

16. 2704 students of a school are arranged in a square for display. Find the number of students in each row.

17. Each member of a cooperative society subscribes 20 times the number of the members in Takas. The total amount raised being Tk. 20480, find the number of members of the society.

18. In a garden 36 trees were left excess while planting 1800 trees in square. Find out the number of trees in each row.

19. What is the least perfect square number which is divisible by 9, 15 and 25 ?

20. Labours were employed to reap paddy from a paddy field. The daily wage of each labour is 10 times of their numbers. If the total daily wage is Tk. 6250, find the number of the labours.

21. The difference of squares of two consecutive numbers is 37. Find the two numbers.

22. Find two such least consective numbers so that the difference of squares of them is a perfect square number.

23. 384 and 2187 are two numbers.

    (a) Verify with factors whether the first number be perfect square number.

    (b) If the second number is not perfect square number, what is the least number to be multiplied to get a perfect square number? what is the perfect square nmuber?

    (c) What is the best number to be added to the second number so that the total sum is a perfect square number?

24. A troops can be arranged in 6,7 and 8 rows, but not is a square form.

    (a) Find out factors of 8.

    (b) What is the least number by which the number in troops is to be multiplied so that the troops can be arranged in a square form?

    (c) At least how many troops should have to join to arrange troops so obtained in a square form?